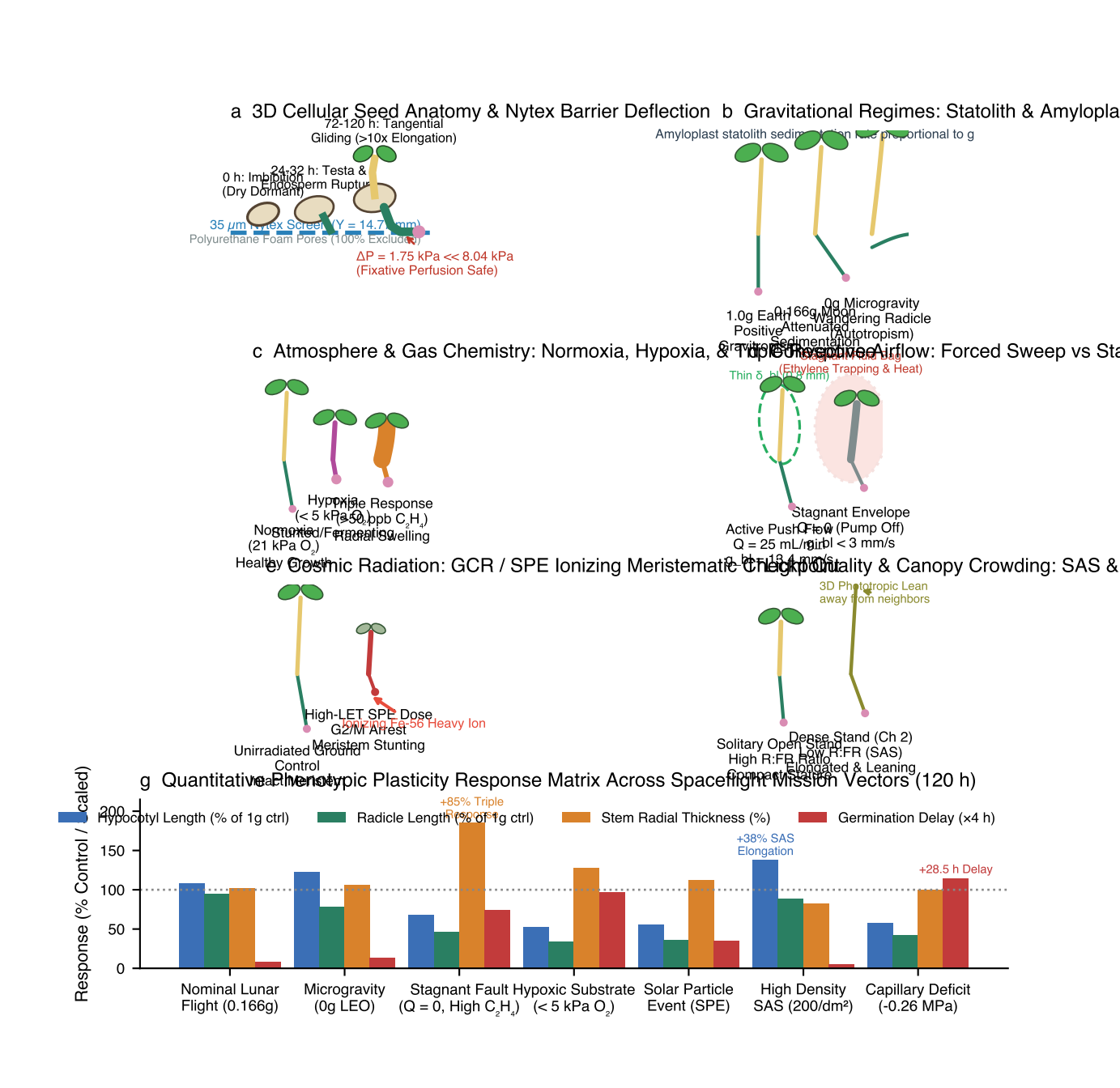
a 3D Cellular Seed Anatomy & Nytex Barrier Deflection
72-120 h: Tangential
Gliding (>10x Elongation)
24-32 h: Testa &
Endosperm Rupture
0 h: Imbibition
(Dry Dormant)
35 μm Nytex Screen (YY = 14.7 mm)
Polyurethane Foam Pores (100% Excluded)
ΔP = 1.75 kPa << 8.04 kPa
(Fixative Perfusion Safe)
b Gravitational Regimes: Statolith & Amyloplast
Amyloplast statolith sedimentation rate proportional to g
1.0g Earth
0.166g Moon
0g Microgravity
Positive
Gravitropism
Attenuated
Sedimentation
Wandering Radicle
(Autotropism)
c Atmosphere & Gas Chemistry: Normoxia, Hypoxia, & Triple Response
Thin δ
(Ethylene Trapping & Heat)
Hypoxia
(< 5 kPa O2)
Triple Response
(~50 ppb C2H4)
Normoxia
(21 kPa O2)
Stunted/Fermenting
Radial Swelling
Healthy Growth
Stagnant Envelope
Active Push-Pull
Q = 25 mL/min
d Cosmic Radiation: GCR / SPE Ionizing
d Microgravity & Canopy Crowding: SAS &
3D Phototropic Lean
away from neighbors
High-LET SPE Dose
Ionizing Fe-56 Heavy Ion
G2/M Arrest
Meristem Stunting
Unirradiated Ground
Control
Dense Stand (Ch 2)
Solitary Open Stand
High R:FR (SAS)
Low R:FR (SAS)
Elongated & Leaning
Compact Status
g Quantitative Phenotypic Plasticity Response Matrix Across Spaceflight Mission Vectors (120 h)
Hypocotyl Length (% of 1g ctrl)
Radicle Length (% of 1g ctrl)
Stem Radial Thickness (%)
Germination Delay (×4 h)
Response (% Control / Scaled)
200
150
100
50
0
+85% Triple
Response
+38% SAS
Elongation
+28.5 h Delay
Nominal Lunar
Flight (0.166g)
Microgravity
(0g LEO)
Stagnant Fault
(Q = 0, High C2H4)
Hypoxic Substrate
(< 5 kPa O2)
Solar Particle
Event (SPE)
High Density
SAS (200/dm²)
Capillary Deficit
(-0.26 MPa)